

Methods 3: Multilevel Statistical Modeling and Machine Learning

Class 03:

What to do when the response variable is not continuous?

September 25, 2025

If you cannot knit a pdf, try html instead
(you can make it a pdf afterwards)

```
---  
title: "Untitled"  
date: "2024-09-19"  
output: html_document  
---
```

```
---  
title: "Untitled"  
date: "2024-09-19"  
output: pdf_document  
---
```

assignment_1, Methods 3, 2025

2025-09-25

REMEMBER: In your portfolio, make sure to include code that can reproduce the answers requested in the exercises below (**MAKE A KNITTED VERSION**). If it does not KNIT, it cannot be part of your portfolio

Exercises and objectives

The objectives of the exercises of this assignment are:

- A) Download and organise the data and model
- B) Fit multilevel models for binomial data
- C) Fit multilevel models for count data

Introduction to experiment



Consciousness and Cognition

Volume 71, May 2019, Pages 59-69



Visual expectations change subjective experience without changing performance

Lau Møller Andersen ^{a b}  , Morten Overgaard ^b, Frank Tong ^c

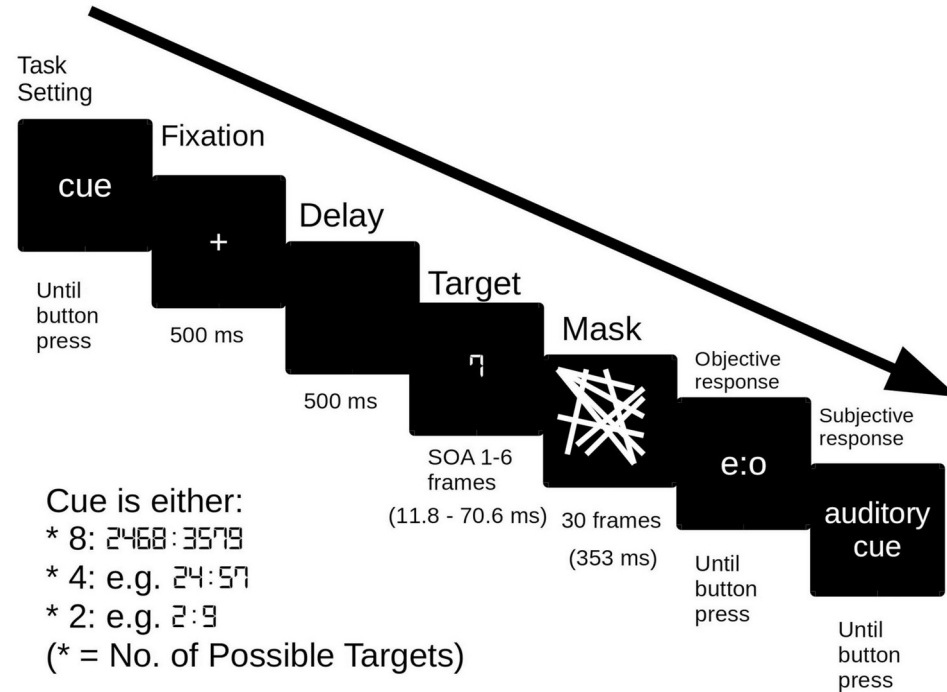
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<https://doi.org/10.1016/j.concog.2019.03.007> 

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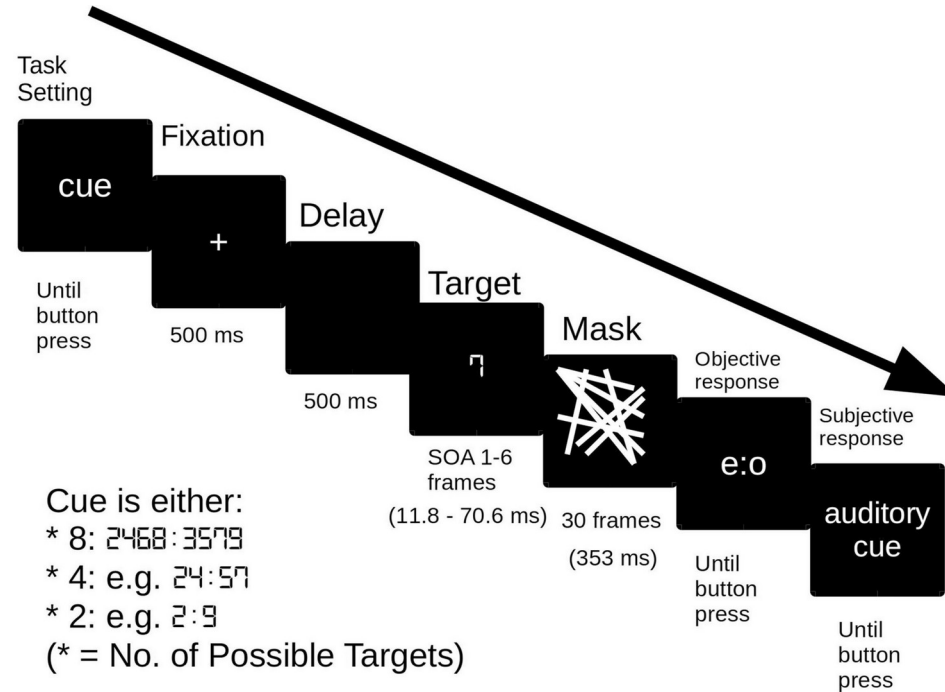
Introduction to experiment



Andersen LM, Overgaard M, Tong F (2019) Visual expectations change subjective experience without changing performance. *Consciousness and Cognition* 71:59–69.
<https://doi.org/10.1016/j.concog.2019.03.007>

Introduction to experiment

We are looking at experiment 2, where Stimulus Onset Asynchrony was always 3 frames, i.e. 35.3 ms



Andersen LM, Overgaard M, Tong F (2019) Visual expectations change subjective experience without changing performance. *Consciousness and Cognition* 71:59–69.
<https://doi.org/10.1016/j.concog.2019.03.007>

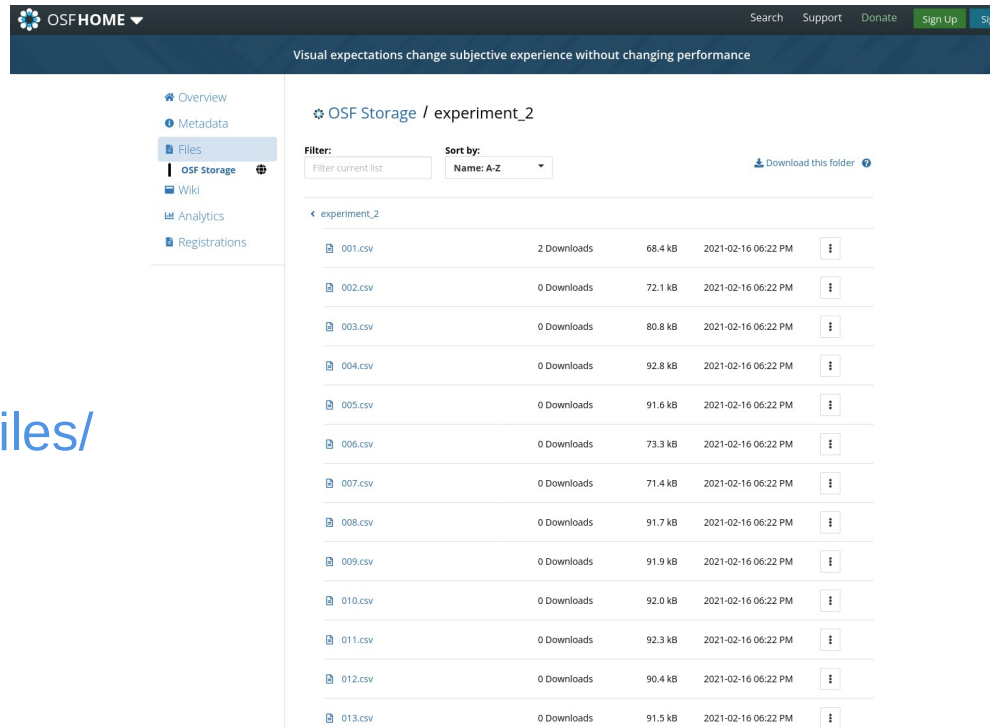
Perceptual Awareness Scale

Table 2. A generalisation of all subjects' description of each category based on the interview following the experiments.

PAS	Category	Description
1	No experience	No impression of the stimulus. All answers are seen as mere guesses
2	Brief glimpse	A feeling that something has been shown. Not characterised by any content, and this cannot be specified any further
3	Almost clear experience	Ambiguous experience of the stimulus. Some stimulus aspects are experienced more vividly than others. A feeling of almost being certain about one's answer
4	Clear experience	Non-ambiguous experience of the stimulus. No doubt in one's answer

Ramsøy TZ, Overgaard M (2004) Introspection and subliminal perception. *Phenomenology and the Cognitive Sciences* 3:1–23.
<https://doi.org/10.1023/B:PHEN.0000041900.30172.e8>

Data – Center for Open Science



The screenshot shows the OSFHOME interface. The top navigation bar includes the OSFHOME logo, a search bar, and links for Support, Donate, Sign Up, and Sign In. A banner below the navigation bar reads "Visual expectations change subjective experience without changing performance". The left sidebar contains a navigation menu with links for Overview, Metadata, Files (selected), OSF Storage, Wiki, Analytics, and Registrations. The main content area displays the "OSF Storage / experiment_2" page. It features a "Filter:" dropdown set to "Filter current list" and a "Sort by:" dropdown set to "Name: A-Z". A "Download this folder" button is visible. Below these controls is a table listing 13 CSV files, each with a download icon, a download count, file size, and timestamp.

File Name	Downloads	Size	Timestamp
001.csv	2 Downloads	68.4 kB	2021-02-16 06:22 PM
002.csv	0 Downloads	72.1 kB	2021-02-16 06:22 PM
003.csv	0 Downloads	80.8 kB	2021-02-16 06:22 PM
004.csv	0 Downloads	92.8 kB	2021-02-16 06:22 PM
005.csv	0 Downloads	91.6 kB	2021-02-16 06:22 PM
006.csv	0 Downloads	73.3 kB	2021-02-16 06:22 PM
007.csv	0 Downloads	71.4 kB	2021-02-16 06:22 PM
008.csv	0 Downloads	91.7 kB	2021-02-16 06:22 PM
009.csv	0 Downloads	91.9 kB	2021-02-16 06:22 PM
010.csv	0 Downloads	92.0 kB	2021-02-16 06:22 PM
011.csv	0 Downloads	92.3 kB	2021-02-16 06:22 PM
012.csv	0 Downloads	90.4 kB	2021-02-16 06:22 PM
013.csv	0 Downloads	91.5 kB	2021-02-16 06:22 PM

<https://osf.io/ecxsj/files/>

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Data contains

- Response times
- Accuracy
- And counts of Perceptual Awareness Scale Ratings

Model comparisons

- In the assignment, you are asked to do model comparisons – on purpose, I ask you to do this before you have had the lecture on quantitative comparisons
 - Model comparisons can also be done without appealing to statistical significance
- If you prefer, you may come back to the assignment in your exam and use the quantitative methods you will learn about performing model comparisons, but it is *not* a requirement

Now let's get to it!